

# ASSIGNMENT - USB IMAGE EXAMINATION

## OVERVIEW

Your firm recently received a work order to process a USB thumb drive belonging to Jessica Langford. Jessica Langford is a former Senior Financial Analyst at Acme Corp. Jessica was fired Thursday, April 3, 2025 for suspicion of mismanaging company resources. Ms. Langford was ordered by the company to turn over her all company resources, including her laptop, by close of business Monday, April 7, 2025. On Wednesday, April 9, 2025 Ms. Langford filed a Qui tam (whistleblower) and wrongful termination lawsuit against Acme Corp. TNT Law Firm represents Acme Corp. and your firm is contracted with TNT to perform digital forensic investigations and recovery.

## INSTRUCTIONS



Case Number: YY-T006

Your firm is tasked with examining the USB thumb drive for possible evidence that Ms. Langford stole company documents after she was fired from her position. You are also hired to recover any important data regarding Ms. Langford’s motives and methods.

Unfortunately, your firm works under a very constrained budget; therefore, the only tools available are FOSS tools like The Sleuthkit and various hex editor like WinHex and xxd. For this assignment, you will examine the disk geometry and content using TSK and a hex editor.

## QUESTIONS

1. Using either a hex editor or TSK, determine the disk partitioning type (MBR or GPT).
2. Using either a hex editor or TSK, how many partitions are on the disk?
3. What is the file system for the first partition?
4. What is the volume label (boot sector or root directory) for this partition?
5. What is the hexadecimal offset for the first partitions root directory? This will require some calculation and verification.
6. What are the first eleven characters of the root directory? Do not give hexadecimal characters and should look familiar.
7. There are three regular files under the root directory of the file system, what are they?
8. Are their any directories under the root directory? Name them.
9. Are these folders allocated?
10. List all unallocated directories?
11. List all unallocated files?
12. Are any unallocated files you just mentions located elsewhere on this disk? List them.
13. Are there any files that may give us insight into the motives or methods of Ms. Langford? List the file names?
14. What is the content of this file?
15. Another team is reporting that Ms. Lanford encrypted her laptop with a password. Can you give them any information that would help their investigation?

## EVIDENCE FILES

|        |                                                                  |
|--------|------------------------------------------------------------------|
| Name   | usb-whistleblower.img                                            |
| Type   | DOS/MBR boot sector; partition 1                                 |
| Size   | 34359738368 bytes                                                |
| MD5    | 850532AE1CA15B95E141EABE8004695A                                 |
| SHA1   | 0CBBB41DCA93351C4D95A597658FD3FC4B7E6375                         |
| SHA256 | 0968EA1DF2F80CA5CFB59611ABE9D4EAC5FD49E8127017AEF0306CFA61FA3AE9 |

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| <b>SSDeep</b>      | 3072:nzm7rZHXCTmS6Frr1m7rZHXCTmS6om7rZHXCTmS6lm7rZHXCTmS6:inZSiVAnZSi5nZSi8nZSi |
| <b>Description</b> | Raw disk image                                                                  |

## REQUIRED TOOLS

| Name       | Vendor        |
|------------|---------------|
| WinHex     | xWays         |
| 010 Editor | SweetScape    |
| TSK        | Brian Carrier |

## REQUIRED FILES

- evidence-report-template.docx

## DELIVERABLE REQUIREMENTS

- Evidence report with the following sections and requirements
  - Executive Summary
  - Evidence Seized
  - Tools Used
  - Case Information, Observables, Investigator Processes
    - Question answers
  - Conclusion
  - Exhibits (Not required)
  - References (Not required)
  - Photos/screenshots, w/ captions, are required to justify your findings
    - In line with your question answers, not in an appendix/exhibit
  - Must use report template

## GRADING RUBRIC

- Evidence report - 100 pts
  - Completed answers - 60 pts
    - Answered questions for each item - 2 pts
    - Provided screenshot for each item - 2 pts
  - Used template - 30 pts
  - Following instructions - 10 pts